FINAL REPORT: HM-02 & HM-05 REMEDIATION AND RE-VERIFICATION

Project: Hikmalayer (hikmalayer-core) **Files in scope:** `src/api/routes.rs`, `src/persistence.rs`, `src/blockchain/transaction.rs`, `src/blockchain/state.rs` **Prepared by:** Placement Student, Bestower Labs Limited **Classification:** Internal — Confidential **Status:** Both findings closed as of this report

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1. Executive Summary

This report consolidates three rounds of work against two findings from the 22 June 2026 Hikmalayer Security Assessment:

- **HM-02 (High)** — Missing authorization on state-changing endpoints.
- **HM-05 (Medium)** — Unencrypted, non-atomic full-state persistence.

Round 1 applied the initial fixes against the baseline codebase (commit bd0bc21): admin-token gating added to five endpoints (plus a sixth found during remediation), and `save_state()` rewritten to an atomic write pattern.

Round 2 re-verified both findings against an updated codebase that had undergone a substantial architectural rewrite (nonce-based signed transactions, a credential registry, VRF-based validator selection, an externally-signed propose/submit mining flow). This found that **HM-05 had fully regressed** — `save_state()` was back to a plain, non-atomic write — and **HM-02 had partially regressed** — `POST /mine` had silently lost its authorization check during the rewrite, while four other endpoints remained correctly gated and three had been redesigned to use per-account cryptographic signatures instead of the shared admin token (flagged as "cannot certify without `transaction.rs`"). Both regressions were fixed in this round.

Round 3 obtained and reviewed `transaction.rs` and `state.rs` to close the open question from Round 2: whether the new signature-based authorization on `/tokens/transfer`, `/staking/deposit`, `/staking/withdraw`, and `/credentials/*` actually enforces sender identity and replay protection, or only checks "is this *a* valid signature." **Verified correct** on all counts, with existing unit tests as supporting evidence — no code changes were required in this round.

Net result: both HM-02 and HM-05 are closed as of the code reviewed in Round 3, with every authorization and persistence claim in this report backed by either a direct code inspection or a passing unit test, not assumption.

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2. HM-02 — Missing Authorization on State-Changing Endpoints

2.1 Original finding (22 June 2026 assessment)

Five state-changing REST handlers — `issue_certificate`, `transfer_tokens`, `mine_block`, `stake_tokens`, `withdraw_stake` — had no authentication or authorization check at all. Any network caller could mint certificates, move tokens, trigger block production, or manipulate validator stake. CWE-862 (Missing Authorization). Indicative severity: High (~8.6 CVSS).

2.2 Round 1 remediation

Each handler (plus `set_mining_difficulty`, found to share the same defect) was updated to accept a `HeaderMap` and reject requests that fail `authorize_admin(&headers, &state)` — the same fail-closed admin-token check already used correctly elsewhere in the codebase (`/governance/config`, `/slashing/evidence`).

2.3 Round 2 re-verification findings

Against the rewritten codebase:

Endpoint	Status found	Action
`/certificates/issue`	Still correctly gated	None needed
`/certificates/attest` *(new)*	Correctly gated	None needed
`/tokens/faucet` *(new)*	Correctly gated	None needed
`POST /mining/difficulty`	Still correctly gated	None needed
`/tokens/transfer`	Redesigned to signed transactions via `queue_transaction()`	Flagged — needed `transaction.rs` to certify
`/staking/deposit`	Redesigned to signed transactions	Flagged — needed `transaction.rs` to certify
`/staking/withdraw`	Redesigned to signed transactions	Flagged — needed `transaction.rs` + `state.rs` to certify
`/credentials/issue`, `/credentials/revoke` *(new)*	Redesigned to signed transactions	Flagged — needed `transaction.rs` to certify
`/slashing/equivocation` *(new)*	Gated by cryptographic proof verification	Correctly designed as-is
`POST /mine`	**Admin-token check missing entirely — regressed**	**Fixed in Round 2**

Fix applied (`routes.rs`):

```
        // Before (regressed):
        async fn mine_block(State(state): State<AppState>) ->
Json<MiningResponse> {
            let finality_depth = { ... };
            ...
        }

        // After (re-fixed):
        async fn mine_block(State(state): State<AppState>, headers:
HeaderMap) -> Json<MiningResponse> {
            if !authorize_admin(&headers, &state) {
                return Json(MiningResponse {
```

```
                    status: "error".to_string(),
                    message: "Unauthorized: a valid admin token is required
to trigger mining"
                        .to_string(),
                    block_index: 0,
                    transactions_count: 0,
                });
            }
            let finality_depth = { ... };
            ...
        }
```

Even though this codebase version only *succeeds* at mining when the node's own locally-held `VALIDATOR_PRIVATE_KEY` is PoS-selected for the slot (so an outsider cannot forge blocks through this route), leaving it unauthenticated still allowed any caller to (a) trigger wasted lock/CPU work on demand and (b) infer exactly when this node is the active validator by watching for success — a targeted-liveness/DoS signal against that validator's slot. Gating it matches how the rest of the codebase treats operator-triggered actions.

2.4 Round 3 — closing the signature-based endpoints

`transaction.rs` and `state.rs` were reviewed against three specific requirements for the redesigned endpoints to be considered properly authorized:

Requirement 1 — the signing key must be bound to the claimed sender, not just "any valid signature."

Confirmed in `transaction.rs`:

```
        fn verify_sender_signature(&self, from: &str, message: &str) ->
Result<(), String> {
            let public_key = self.public_key.as_ref().ok_or_else(...)?;
            let signature = self.signature.as_ref().ok_or_else(...)?;
            let derived = pos::derive_address(public_key)?;
            if derived != *from {
                return Err("Sender address does not match the signing
key".to_string());
            }
            if !pos::verify_message(message, public_key, signature) {
                return Err("Transaction signature verification
failed".to_string());
            }
            Ok(())
        }
```

Backed by an existing test that proves the exact attack this guards against (attacker signs with their own key but claims to act on the victim's address) is rejected:

```
        fn transfer_rejects_key_not_matching_sender() { ... assert!
(tx.verify_for_block(...).is_err()); }
```

Applies to `Transfer`, `Stake`, and `Certificate`/credential transactions. **Verified.**

Requirement 2 — `Withdraw` must verify against the on-chain registered key, not a client-supplied one.

`transaction.rs`'s stateless `verify_for_block` deliberately does *not* cryptographically verify `Withdraw` signatures (only checks one is present) — by design, deferring to the stateful layer. Confirmed in `state.rs::apply_transaction`:

```
        TransactionType::Withdraw => {
```

```
            ...
            let info = self.stakers.get(from).ok_or_else(...)?.clone();
            let message = Transaction::withdraw_signing_message(from,
tx.amount, tx.nonce);
            if !pos::verify_message(&message, &info.public_key, signature) {
                return Err("Withdraw signature does not match registered
key".to_string());
            }
            ...
        }
```

This verifies against `info.public_key` — the key already registered on-chain for that staker — not anything supplied in the request, which is a tighter design than re-supplying a key (an attacker can't substitute a different key at withdrawal time). Backed by a direct test:

```
        fn withdraw_rejects_wrong_key() { ... assert!
(state.apply_transaction(&withdraw).is_err()); }
```

Verified.

Requirement 3 — nonces must be strictly enforced to prevent replay.

Confirmed in `state.rs`:

```
        fn consume_nonce(&mut self, account: &str, nonce: u64) -> Result<(),
String> {
            let expected = self.nonce_of(account) + 1;
            if nonce != expected {
                return Err(format!("Invalid nonce for {}: expected {}, got
{}", account, expected, nonce));
            }
            self.nonces.insert(account.to_string(), nonce);
            Ok(())
        }
```

This is strict sequential enforcement (`current + 1`, not merely "higher than before"), called for every state-changing transaction type. Backed by:

```
        fn transfer_updates_balances_and_nonce() {
            ...
            // Same nonce cannot apply twice.
            assert!(state.apply_transaction(&tx).is_err());
        }
```

Verified across `Transfer`, `Stake`, `Withdraw`, and `Certificate`/credential transactions.

Bonus check — credential revocation is scoped to the original issuer, not just any signed request:

```
        if record.issuer != *issuer {
            return Err("Only the issuer can revoke a
credential".to_string());
        }
```

Verified.

2.5 HM-02 — Final Status

Endpoint	Final verdict
`/certificates/issue`, `/certificates/attest`, `/tokens/faucet`,	✅ Closed — admin-token gated, unchanged across rounds

`/mining/difficulty`	
`/tokens/transfer`, `/staking/deposit`, `/credentials/issue`, `/credentials/revoke`	✅ Closed — per-account signature verification confirmed correct (Round 3)
`/staking/withdraw`	✅ Closed — on-chain-registered-key verification confirmed correct (Round 3)
`/mine`	✅ Closed — admin-token gate restored (Round 2)
`/slashing/equivocation`	✅ Closed — cryptographic proof verification, correct by design

HM-02 is fully closed.

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3. HM-05 — Unencrypted, Non-Atomic Full-State Persistence

3.1 Original finding (22 June 2026 assessment)

`save_state()` wrote the entire chain state to `data/state.json` via a plain `fs::write` — not atomic, so a crash mid-write could corrupt the whole file — with default (non-restrictive) file permissions. CWE-312 / CWE-311. Indicative severity: Medium.

3.2 Round 1 remediation

`save_state()` rewritten to: create a uniquely-named temp file in the same directory as the target, set `0600` permissions before writing any bytes, `sync_all()` to force the write to durable storage, atomically `rename()` into place, and clean up the temp file on any failure.

3.3 Round 2 re-verification findings

Against the rewritten codebase, `save_state()` had reverted to the exact vulnerable pattern:

```
        // Found in Round 2 (regressed):
        pub fn save_state(snapshot: &AppSnapshot) -> std::io::Result<()> {
            let path = state_path();
            if let Some(parent) = Path::new(&path).parent() {
                fs::create_dir_all(parent)?;
            }
            let data = serde_json::to_string_pretty(snapshot)
                .map_err(|err|
std::io::Error::new(std::io::ErrorKind::Other, err))?;
            fs::write(path, data)
        }
```

No temp file, no `sync_all()`, no permission lock-down. The only substantive change from the original vulnerable version was that the path is now resolved via a `HIKMALAYER_STATE_PATH` environment variable — unrelated to the vulnerability itself. Most likely cause: this file was edited independently during the broader rewrite (e.g. to strip `balances`/`stakers`/nonces out of the persisted snapshot, now correctly treated as pure chain state rebuilt at startup) without the atomicity fix being carried forward.

Fix re-applied (`persistence.rs`):

```
        pub fn save_state(snapshot: &AppSnapshot) -> std::io::Result<()> {
```

```
            let path_string = state_path();
            let path = Path::new(&path_string);
            let parent = path.parent().unwrap_or_else(|| Path::new("."));
            fs::create_dir_all(parent)?;

            let data = serde_json::to_string_pretty(snapshot)
                .map_err(|err|
std::io::Error::new(std::io::ErrorKind::Other, err))?;

            let file_name = path.file_name().and_then(|n|
n.to_str()).unwrap_or("state.json");
            let temp_path = parent.join(format!(".{}.tmp.{}", file_name,
uuid::Uuid::new_v4()));

            let write_result = (|| -> std::io::Result<()> {
                let mut file = File::create(&temp_path)?;
                #[cfg(unix)]
                file.set_permissions(fs::Permissions::from_mode(0o600))?;
                file.write_all(data.as_bytes())?;
                file.sync_all()?;
                Ok(())
            })();

            if let Err(err) = write_result {
                let _ = fs::remove_file(&temp_path);
                return Err(err);
            }
            if let Err(err) = fs::rename(&temp_path, path) {
                let _ = fs::remove_file(&temp_path);
                return Err(err);
            }

            #[cfg(unix)]
            fs::set_permissions(path, fs::Permissions::from_mode(0o600))?;
            Ok(())
        }
```

Functionally identical to the Round 1 fix, adapted to call `state_path()` (the env-var-aware resolver) instead of a hardcoded constant, and to work with the current, slimmer `AppSnapshot`.

3.4 HM-05 — Final Status

HM-05 is closed as of the code delivered in Round 2. No further changes were made or needed in Round 3 (`transaction.rs`/`state.rs` do not touch persistence).

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4. Files Changed (cumulative, across all rounds)

File	Changed?	Summary
`src/api/routes.rs`	Yes	HM-02: 6 endpoints gated in Round 1; `/mine` re-gated in Round 2
`src/persistence.rs`	Yes	HM-05: atomic write applied in Round 1; re-applied in Round 2

`src/blockchain/transaction.rs`	No	Reviewed in Round 3 — verified correct, no changes needed
`src/blockchain/state.rs`	No	Reviewed in Round 3 — verified correct, no changes needed

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5. Verification & Testing Recommendations

HM-02

#	Test	Expected Result
1	`POST /mine` with no/invalid `x-admin-token`	Rejected; no block produced
2	`POST /mine` with valid `x-admin-token`, node PoS-selected	Block mined and gossiped as before
3	`POST /tokens/transfer` with a valid signature from account A but `from: B`	Rejected (`transfer_rejects_key_not_matching_sender` already proves this at the unit level — repeat at the API level for end-to-end confidence)
4	`POST /staking/withdraw` signed with a key other than the one registered on-chain for that staker	Rejected (`withdraw_rejects_wrong_key` proves this at the unit level)
5	Replay a previously-mined transaction verbatim (same nonce)	Rejected (`transfer_updates_balances_and_nonce` proves this at the unit level)
6	Regression: `/certificates/issue`, `/certificates/attest`, `/tokens/faucet`, `/mining/difficulty`	Continue to require the admin token, unaffected by this round's changes

HM-05

#	Test	Expected Result
1	Inspect `data/state.json` permissions after a save (Unix)	`-rw-------` (`0600`)
2	Kill the process mid-write (delay injected before `rename`)	Previous valid state file intact; no leftover `.tmp.*` files after next clean save
3	Point `HIKMALAYER_STATE_PATH`	`save_state()` returns `Err`; no `.tmp.*`

	at a read-only directory	file left behind
4	Concurrent rapid mutating API calls	Each save uses a UUID-suffixed temp file; no collisions

Standing recommendation (raised in Round 2, still open)

Both regressions in Round 2 happened because authorization and persistence hardening are applied **per-handler / per-function, opt-in**, with nothing structural to catch a missing check during a refactor. Recommend:

- A router-level test that enumerates every `POST`/`PUT`/`DELETE` route and asserts each is either admin-gated or on an explicit allowlist of signature-verified routes.
- A persistence-level test that asserts `save_state()` never writes directly to the final path (e.g. briefly observe a `.tmp.*` file, or assert `0600` immediately after a save).

Neither exists yet; both would have caught this round's regressions automatically instead of requiring manual re-review.

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6. Sign-off

Finding	Status	Verified by
HM-02 — Missing authorization on state-changing endpoints	**Closed**	Code review + existing unit tests (`transfer_rejects_key_not_matching_sender`, `withdraw_rejects_wrong_key`, `transfer_updates_balances_and_nonce`)
HM-05 — Unencrypted, non-atomic full-state persistence	**Closed**	Code review (atomic write pattern re-applied and inspected)

Remaining findings from the original 22 June 2026 assessment (HM-01 key custody, HM-03 JWT/session verification for non-signed-transaction routes, HM-06 secret rotation, HM-07 constant-time comparison, HM-08 consensus/Sybil hardening) are outside the scope of this report and remain tracked separately.